LOKESH RAM HARIRAM

(512)7018465 | lokeshramloka@gmail.com | LinkedIn | GitHub

SKILLS

AI/ML & LLM Engineering: Python, LLMs, Generative AI, OpenAI API, Claude API, RAG, AI Agents, Embeddings, Vector Search, Tool Calling, Structured Outputs, LLM Evaluation, Guardrails

AI Application Engineering: Natural Language-to-SQL, RAG Pipelines, Agentic Workflows, Human-in-the-Loop Systems, Model Evaluation, FastAPI, REST APIs, JSON Schema, API Integration, Contract Validation

ML & Data Science: XGBoost, ARIMA, Predictive Modeling, Feature Engineering, Time-Series Forecasting, Statistical Analysis, Exploratory Data Analysis

Cloud & Production Engineering: AWS, Azure, Azure Data Factory, Azure Synapse Analytics, Docker, Kubernetes, CI/CD, ETL/ELT, Data Modeling, Data Transformation, PostgreSQL

Analytics & Visualization: Power BI, DAX, Power Query, KPI Analysis, Dashboard Development

Engineering & Collaboration: Git, GitHub, Requirements Gathering, Technical Discovery, Stakeholder Collaboration, Technical Prototyping, Systems Integration, Solution Design

PROFESSIONAL EXPERIENCE

Applied AI/ML Engineer — Aug 2025 – Jan 2026

American College of Emergency Physicians (ACEP), Irving, TX

- **Built AskEMDI**, an LLM-powered natural-language-to-SQL platform using Amazon Bedrock and Anthropic Claude, enabling emergency physicians and researchers to query structured emergency-medicine data without writing SQL.
- **Engineered** ETL/ELT pipelines using Azure Data Factory and Azure Synapse Analytics to ingest, cleanse, transform, and validate a **23M+** encounter dataset of de-identified clinical data, creating the structured data foundation for AskEMDI and research analysis.
- **Built** Python and SQL data-quality validation workflows to identify missing values, inconsistent records, and clinical coding issues before data reached downstream LLM and analytics workflows, improving data reliability for AI-driven analysis.
- **Partnered** with clinicians and operations stakeholders to gather requirements, demonstrate prototypes, and iterate on query logic and outputs, translating real-world research questions into usable AI workflows for non-technical physicians and researchers.

Research Assistant (Machine Learning) — Aug 2024 – May 2025

SEAR LAB, Arlington, TX

- **Built** Python and SQL data pipelines for **500K+** time-series sensor records, covering data cleaning, feature engineering, and transformation to produce model-ready datasets for predictive modeling and operational analysis.
- **Developed XGBoost** and **ARIMA** models to analyze sensor data, identify drivers of operational performance, and generate predictive insights that contributed to a **15%** improvement in production efficiency within the research environment.
- **Automated** data workflows and developed Python/Matplotlib visualizations and Power BI dashboards, reducing KPI reporting time by **40%** and improving access to operational insights.
- **Partnered** with research and engineering stakeholders to define KPI requirements, demoed prototype dashboards, and iterated on features and reporting from their feedback, with final dashboards adopted for weekly KPI tracking.

EDUCATION

The University of Texas at Arlington — May 2026
Master of Science in Information Systems

Sathyabama University — May 2024
Bachelor of Science in Computer Science

PROJECT EXPERIENCE

Longitudinal Meeting Intelligence & Accountability System

Tech Stack: Python, Claude API, Embeddings, PostgreSQL, FastAPI, Docker

- Built a longitudinal meeting-intelligence system using publicly available city council meeting minutes, extracting action items, decisions, owners, deadlines, and issue context from recurring sessions and persisting them for cross-meeting tracking.
- Engineered cross-meeting entity resolution using embedding similarity followed by Claude-based verification to distinguish newly raised issues from semantically related but distinct historical action items.
- Developed temporal state tracking to identify overdue, stale, and repeatedly resurfacing issues, maintaining an evidence trail across meetings to surface unresolved commitments rather than treating each meeting as an isolated document.
- Built an interactive accountability dashboard and created a manually labeled evaluation set of 75 meeting/action-item examples to assess action-item extraction and cross-meeting matching quality, analyzing false matches and missed links to identify recurring sources of model error.

E-Commerce Pricing Risk Detection System

Tech Stack: Python, XGBoost, Sentence Transformers, PostgreSQL, FastAPI, Streamlit, Docker

- Built a pricing-risk detection system using the Amazon Reviews 2023 raw_meta_Electronics dataset to identify potentially erroneous or suspicious product prices using product descriptions, category context, features, and price information.
- Engineered categorical, price-distribution, and semantic features including category-level price deviation, relative discount magnitude, product attributes, and Sentence Transformer embeddings of product titles and descriptions, using XGBoost to generate product-level pricing-risk scores.
- Developed a FastAPI scoring service and Streamlit investigation dashboard that prioritizes high-risk products for human review, displaying comparable category pricing, product attributes, contributing signals, and model explanations for each flagged listing.
- Created a labeled evaluation benchmark by injecting realistic pricing anomalies into genuine product-price records and tuned detection thresholds using precision, recall, false-negative cost, and manual-review volume to balance financial risk against investigation workload.

API Integration & Contract Verification Agent

Tech Stack: Python, Claude API, Shippo API, FastAPI, Docker, JSON Schema

- Built an API integration agent using Claude to parse Shippo documentation for 10 selected endpoints, generating endpoint clients, request mappings, response schemas, and executable integration tests from documented API contracts.
- Implemented a bounded generate–execute–diagnose–repair workflow that runs generated requests against the Shippo test environment, validates responses against expected schemas, and retries once for defined failures such as missing required fields, incorrect parameter mappings, and invalid request formats.
- Engineered contract-validation workflows to compare documented request and response structures with observed API behavior, identifying schema mismatches, unexpected response fields, missing parameters, and inconsistent error responses across tested endpoints.
- Developed an endpoint-level verification report that records test outcomes, detected discrepancies, repair attempts, unresolved failures, and human-review requirements, providing engineers with an auditable integration handoff for unfamiliar third-party APIs.